

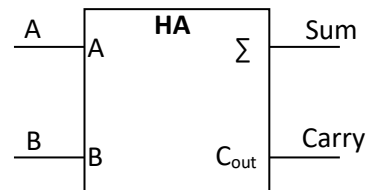
Objectives:

- To study the characteristics of a half adder and a full adder
- To implement a four bit parallel adder
- To implement BCD to seven-segment decoder
- To study the basic operation and design of a decoder/multiplexer circuit
- To describe the concept of active –low and active-high logic signals

Half Adder

Addition is the most commonly performed arithmetic operation in digital systems. To perform addition operations, two types of adders can be used: half adder (HA) or full adder (FA). The two types of adder are going to be discussed during this lab session. The half adder accepts two binary digits on its inputs and produces two binary digits on its outputs, a sum bit (Sum) and a carry bit (Carry). The following truth table represents the results of a half adder (HA).

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Above table shows that the Sum function is the result of Exclusive ORing the inputs A and B, while the Carry function is the result of ANDing the inputs A and B. Then,

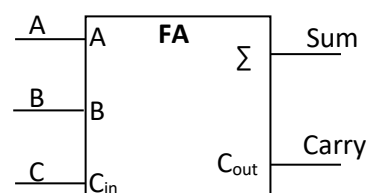
$$Sum = A \oplus B \quad Carry = AB$$

Full Adder

The full adder accepts two input bits A and B and an input carry C and generates a sum output (Sum) and output carry (Cout). The following truth table represents the results of a full adder (FA).

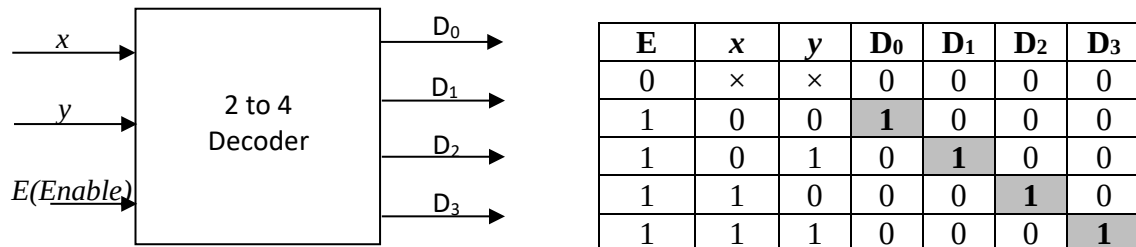
$$Sum = A \oplus B \oplus C \quad Carry = AB + (A \oplus B).C$$

A	B	C	Sum	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Decoders

A decoder is a combinational circuit that converts coded inputs to another coded outputs. The famous examples of decoders are binary n-to- 2^n decoders and seven-segment decoders. A binary decoder has n inputs and a maximum of 2^n outputs. As we know, an n-bit binary number provides 2^n minterms (sum of products) or maxterms (product of sums). This type of decoder produces one of the 2^n minterms or maxterms at the outputs based on the input combinations. Lets take the 2-to-4 decoder as an example, the block diagram and the truth table of this decoder is shown below.

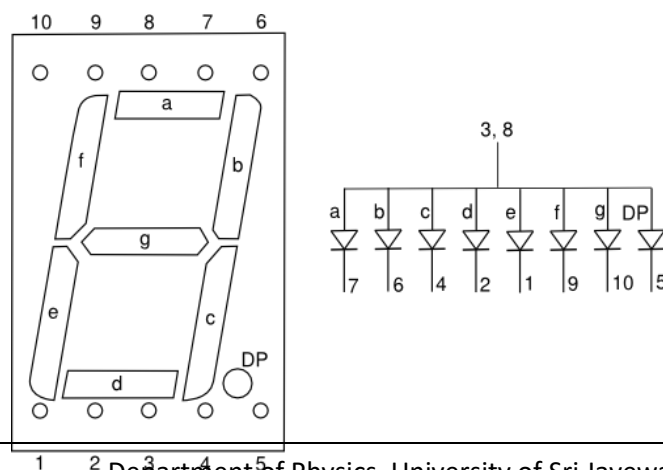


From the truth table, you can observe the basic operation of n-to- 2^n decoders, there is only one active output (minterm) for each input combination. The Boolean expression of the output signals are:

$$D_0 = E\bar{x}\bar{y} \quad D_1 = E\bar{x}y \quad D_2 = Ex\bar{y} \quad D_3 = Exy$$

BCD to Seven-Segment Decoder

Another common type of decoder is the seven-segment decoder. This decoder is used along with seven-segment LED display to create decimal or hexadecimal digits. The Seven segment LED display is commonly used for numerical displays as in multimeters and calculators, it contains seven independent LEDs arranged as below figure.



Common Anode Seven Segment Display

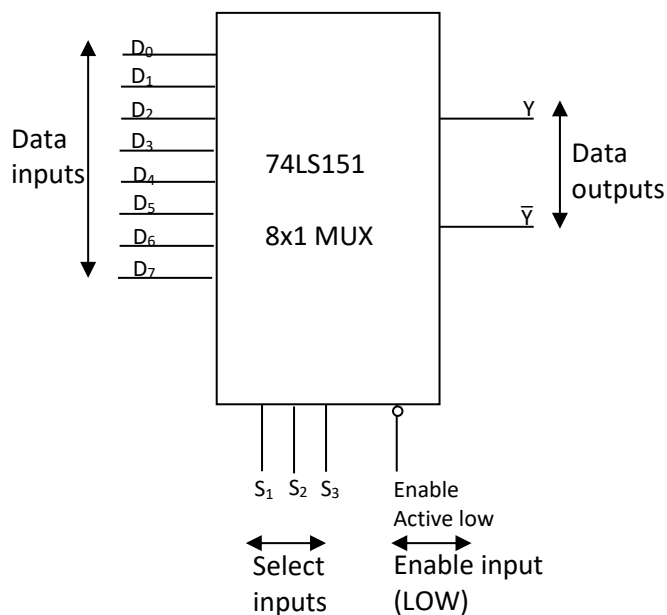
There are two main types of seven-segment LEDs, the common cathode (CC) and the common anode (CA). In the CC type, the cathodes for all segments are joined in a single node. On the other hand in CA type, the anodes are joined together in a single node.

All decimal or hexadecimal numbers can be displayed by controlling the state of the appropriate segments ON or OFF. This can be done using a seven-segment decoder, a seven-segment decoder accepts four binary inputs and provides seven outputs that determine which of the segments on a seven-segment LED display should be on or off to create a decimal or hexadecimal digits.

As an example of the commercial 7-segment decoder is the 7447 chip. This is a BCD to seven-segment decoder which used for displaying the numbers from 0 through 9 based on the corresponding input BCD number. This chip is design for use with to a common anode seven segment display. It has active-low outputs (see Pin-Out Diagram and Function Table).

Multiplexers (Data Selectors)

A digital multiplexer is a binary controlled digital signal switch with many inputs, and one output selected by several digital control lines. A 74LS151 has eight inputs that can be individually selected by three select lines. The output is connected to the input line selected by the binary value on the three select lines. If the three select lines are all zeros, then input line "0" is selected and connected to the output line.



Activity 01 – The Half Adder

1. Construct the circuit as shown in Figure 1.

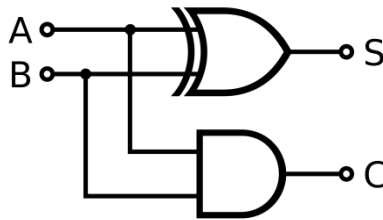


Figure 1. The Half adder circuit

2. Vary the inputs A and B (i.e. 0 and +5V) to obtain all the possible combinations and complete a Truth table for the sum output S and the carry output C.
3. Is this as you expected? Why is the half-adder not suitable for all addition operations? (i.e. why do we need a full adder?)

Note: Handling ICs - Remember to always turn off the power on your breadboard before installing or removing an integrated circuit (IC). Also, please use the available tools to install and remove the IC circuits. Doing this by hand will very often bend the pins.

Activity 02 – The Four-bit Adder

1. The 74LS83a is a high-speed 4-bit Full Adder. It accepts two 4-bit words (A_1 to A_4) and (B_1 to B_4) and a carry input (C_0). It generates the binary sum outputs (S_1 to S_4) and the carry output (C_4) from the most significant bit. Connect the two binary numbers shown below to the inputs of the 4-bit parallel adder. Connect the carry input C_0 to 0.

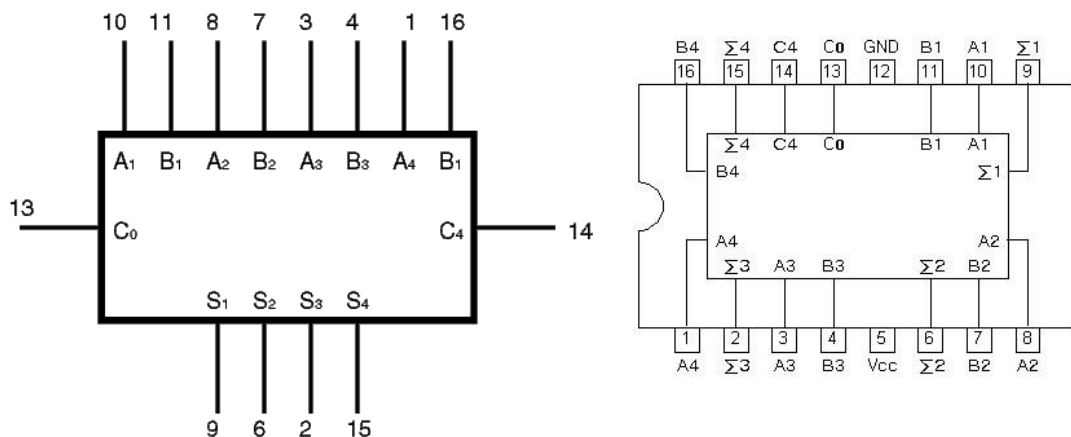
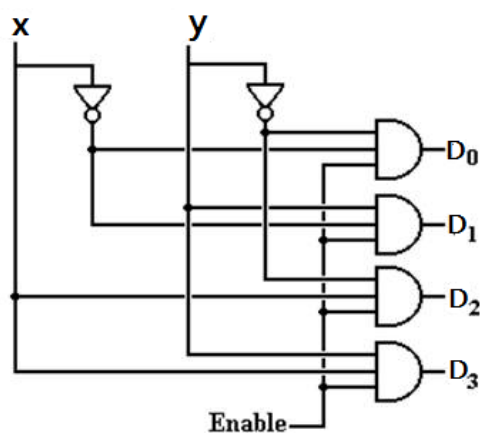


Figure 2. 74LS83 the Four Bit Adder

- Check the operation of the Four Bit adder by adding the numbers:
(a) $0110 + 0101$ (b) $1111 + 1111$
- What is the value of the addition (a) when $C_0 = 1$?
- What is the value of the addition (b) when $C_0 = 1$?

Activity 03 – Two to Four Decoder

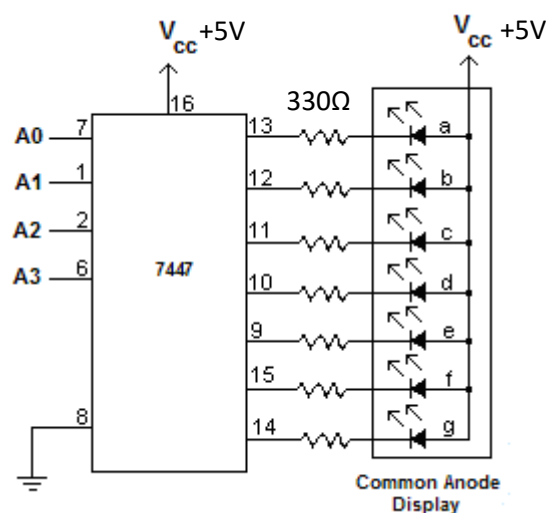
- Construct the logic circuit of the 2-to-4 Decoder that is shown in the following figure. (Use 74LS11 triple 3-input AND logic IC)
- Try all input combinations and fill in the following truth table:



E	x	y	D ₀	D ₁	D ₂	D ₃
0	x	x				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

Activity 04 – Seven Segment Display

- Construct a BCD to SSD decoder circuit using 74LS47 as shown below. Use 8, 330 Ω resistors between SSD display and IC.

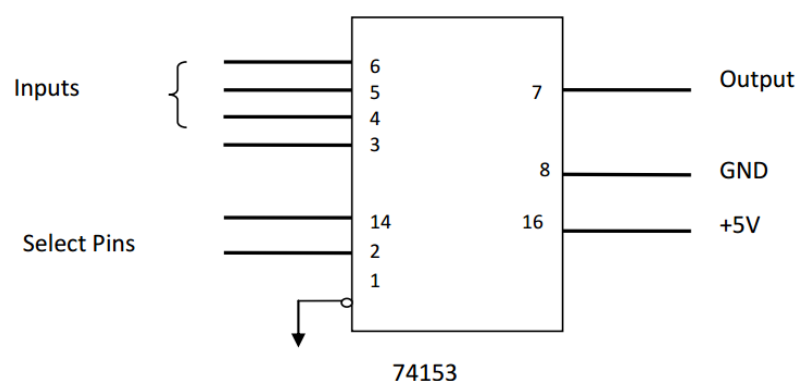
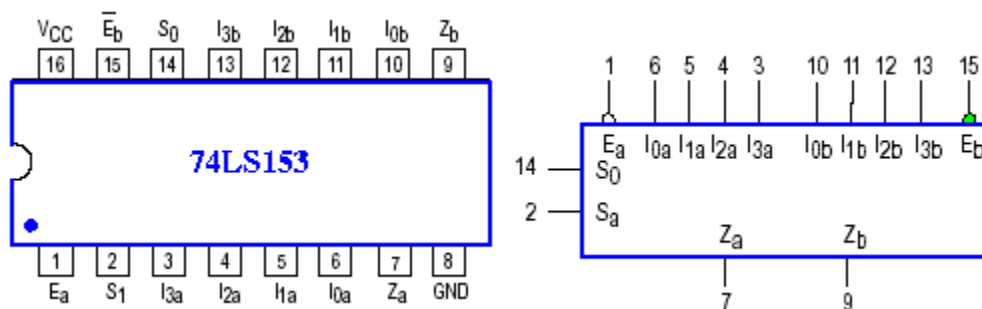


- Verify the function of the circuit by applying different BCD numbers and monitoring the corresponding decimal digits on the 7-segment LED display. (pin 3, 4, and 5 in 74LS47 should be connected to $V_{CC} +5V$, see the data sheet for more information)

Decimal or Function	A ₃	A ₂	A ₁	A ₀	a	b	c	d	e	f	g
0											
1											
2											
3											
4											
5											
6											
7											
8											
9											

- Construct truth tables for all seven circuits. Draw Karnaugh maps and write logic expressions for each of them. Draw a circuit diagram for any of them. (Post laboratory work)
- Connect 74LS47 decoder to 74LS83 parallel adder such that you can see the output in the SSD.

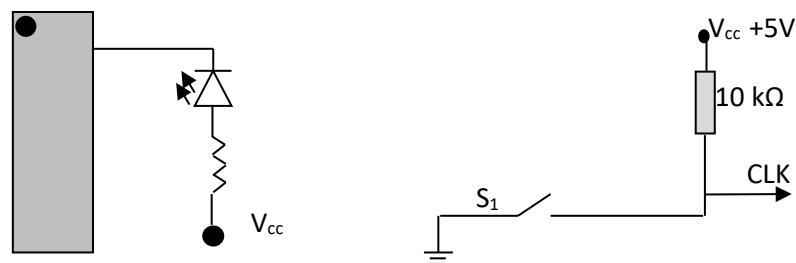
Activity 05- Data Selector



The TTL IC 74LS153 is a Dual 4- Line to 1- line multiplexer. By setting a two-bit address (S0 and S1) one of the four inputs can be connected to the single output line.

Connect one of the inputs of the 74LS153 to the pulse generator as shown above and observe/record how the inputs can be selected by setting an address.

Note: The “ON” state of LED is defined to be the “0” state and vice versa. The LED is connected in reversed biased condition in order to protect the IC.



In case to provide clock signal or change logic states (Low and High) you can make a de-bounced switch as shown above.

